Advanced AI-Driven Cloud & DevSecOps Capstone Project

Participant Guide

1. Customer Background

Acme Retail Ltd. is modernizing its engineering organization through an AI-Driven Cloud & DevSecOps Modernization Program. A new Inventory Management System (IMS) is ready for deployment, but engineering processes are inconsistent, manual, and difficult to scale.

2. Your Role

Platform Engineer

3. Business Problems

- Duplicate pipelines
- Duplicate Terraform
- Different repository structures
- Inconsistent AI specifications
- Long onboarding
- High maintenance

4. Your Mission

As the **Platform Engineer**, your responsibility is to transform the engineering solution developed for a single application into reusable platform capabilities that can be adopted consistently across multiple application teams.

You are expected to approach this Capstone Project as a real platform engineering engagement, focusing on standardization, reusability, automation, and developer experience rather than solving the problem for a single application.

Your mission is to:

- Analyze common engineering challenges faced by multiple development teams.
- Identify opportunities to eliminate duplication through reusable engineering assets and standardized practices.
- Design reusable platform capabilities that simplify application onboarding and delivery.
- Create standardized AI Engineering Specification (.md) files that can be reused across projects.
- Develop reusable GitHub Actions workflows, Infrastructure as Code modules, repository templates, and engineering standards.
- Establish consistent approaches for CI/CD, security, testing, documentation, and automation.
- Improve developer productivity by enabling self-service platform capabilities and standardized engineering workflows.
- Validate that the reusable platform components can support multiple application teams with minimal customization.
- Document platform architecture, engineering standards, reusable components, implementation guidelines, and design decisions.
- Deliver a scalable Internal Developer Platform that enables secure, consistent, and efficient software delivery across multiple application teams.

Expected Outcome

- Standardized engineering practices across multiple teams
- Reusable platform components and automation
- Improved developer experience and onboarding
- Secure, scalable, and maintainable platform engineering solutions
- Comprehensive documentation and engineering standards
- A platform foundation that accelerates software delivery across the organization

5. Expected Engineering Approach

Follow this workflow:

1. Understand the business problem.
2. Break the solution into engineering domains.
3. Create AI Engineering Specification (.md) files.
4. Use an AI coding agent to generate an initial implementation.
5. Review, validate, refactor and secure the generated solution.
6. Test, document and package the final solution.

6. Required AI Engineering Specifications

- platform-spec.md
- terraform-modules-spec.md
- workflow-spec.md
- repo-template-spec.md
- governance-spec.md
- developer-experience-spec.md

7. Recommended AI Tools

You may use any AI coding assistant, including:

- GitHub Copilot Agent
- Claude Code
- Cursor
- OpenAI Codex
- Amazon Q Developer
- Gemini CLI

Choose the tool(s) you are most comfortable with. The evaluation focuses on the quality of the specifications and final engineering solution, not on the specific AI tool used.

8. Recommended Technology Stack

Version Control: Git & GitHub

CI/CD: GitHub Actions

Infrastructure as Code: Terraform

Containers: Docker

Cloud: AWS (preferred) or Azure

Security: Trivy, Checkov, Gitleaks

Documentation: Markdown, Mermaid, Draw.io

Presentation: PowerPoint or PDF

9. Repository Naming Convention

To maintain consistency across all submissions, participants must create their GitHub repository using the following naming convention.

Repository Name Format
<group_number>-<capstone_level>

- group_number – Your Assigned Group Number
- capstone_level – foundation, advanced or expert

Examples
group1-foundation
group7-advanced
group12-expert

Please ensure the repository name follows this format exactly, as it will be used during repository provisioning, submission tracking and evaluation.

10. Tools & Development Environment

Provided

- GitHub Repository
- GitHub Actions (if available)
- GitHub Copilot

Recommended Tools

- Terraform
- Docker
- Kubernetes (Kind, Minikube or K3d)
- Trivy
- Checkov
- Gitleaks
- Markdown
- Mermaid
- Draw.io
- Open-source testing frameworks

You may use any AI coding assistant such as GitHub Copilot, Claude Code, Cursor, OpenAI Codex, Amazon Q Developer or Gemini CLI.

Important Notes

- Cloud infrastructure may not be provided. Design, develop and validate your solution locally using Docker, local Kubernetes clusters, LocalStack (where applicable), emulators or other suitable open-source alternatives.
- If GitHub Actions is available, validate your CI/CD workflows using GitHub Actions.
- If GitHub Actions is not available, create the workflow YAML files under .github/workflows/ following GitHub Actions best practices.
- The submitted solution should be reproducible without requiring dedicated cloud resources.

11. Repository Structure

README.md
app/
infrastructure/
.github/workflows/
docs/
architecture/
ai-specifications/
engineering-decisions/
presentation/

12. Engineering Decisions

Document important architectural decisions using ADR format (Context, Problem, Decision, Alternatives, Trade-offs, Consequences, Rationale).

13. Deliverables

- GitHub repository
- AI Engineering Specification files
- Generated and validated implementation
- Infrastructure as Code
- CI/CD workflows
- Architecture documentation
- Engineering Decisions
- Final presentation

14. Success Criteria

Reviewers assess:

- Understanding of the business problem
- Quality of AI Engineering Specifications
- Engineering quality of the final solution
- Security and automation
- Validation of AI-generated artifacts
- Documentation and communication

15. Important Rules

- Do not submit AI prompts or chat history.
- The AI Engineering Specification is the primary artifact.
- You are responsible for validating every AI-generated output before submission.